

Welcome to 151B: Deep Learning

Instructor: Annapurna Vadaparty

TA: Ben Bao

Tutors: Alexander Radulescu, Trevor Duong

Seating

Podium

Zone 1:	Zone 2:	Desks Row 1	Zone 3:	Zone 3:
<names>	<names>		<names>	<names>
		Desks Row 2		
Empty row				
Empty row				
Empty row				

1. Introduce yourself to the other people in your zone
2. Fill out the pre-course survey if you haven't already

Annapurna Vadaparty



How do we think? As a reductionist—I wanted cells!

- B.S. Neuroscience - JHU, 2014-2018

How does *anything* think? Information processing...

- B.S. Computer Science - JHU, 2014-2018

Teaching was always fun—I TA-ed and tutored a lot



I want to do computational neuroscience, but my research is a bit scattered...I'll get a job and email around for research!

- Software Engineer - Oracle 2018-2020

Annapurna Vadaparty



Computational Neuroscience—modeling brains with neural nets

- Research assistant - Columbia, 2020
-



Cognitive Psychology—how do people solve problems?

- Research assistant - Stanford, 2021
 - Jay McClelland will be giving a guest lecture on Jul 20!
-



UC San Diego

Cognitive Science—how does the “click” happen for students?

- Graduate student - UCSD Cognitive Science, 2022-24

Computer Science—CS Education

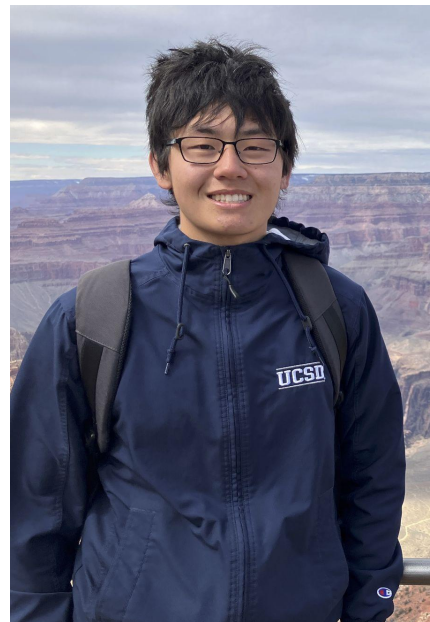
- Graduate student - UCSD Cognitive Science, 2025-26

Ben Bao

I am a first-year Master's student in Computer Science. My academic focus is on machine learning and general software development. I hope to be able to apply my skills towards technologies that improve the lives of people and help them live more fulfilling lives. I did my undergraduate here at UCSD in Cognitive Science, so feel free to chat with me about anything brain or psychology related!

In my spare time I enjoy cooking, guitar, hiking, backpacking, and playing video games.

Email: tibao@ucsd.edu



Trevor Duong

I am an incoming 4th year computer science major at UCSD. My academic and project interests focus heavily on software engineering and machine learning implementations. I am really excited to assist with your learning in any way I can!

For fun I love weightlifting, playing the piano/drums, playing video games, and snowboarding!

Email: trduong@ucsd.edu



Alexander Radulescu

I am an incoming 3rd year computer science major here at UCSD. Academically I am focused on ML and Systems. I am very excited to meet all of you this quarter and help answer any questions you have! For fun I am often swimming or weightlifting, and love to make copious amounts of coffee in my spare time.

Email: alradulescu@ucsd.edu



Today

- Any **blockers** so far?
 - Pre-course survey
 - DSMLP & Colab setup
 - PrairieLearn & PrairieTest Setup: **CBTF Practice exam is mandatory - Ends Jul 2**
 - Accessing any course material
 - Communicate with me about accommodations if you have any
- **Goals & philosophy** of the course
- **Standards-based grading:** High Pass, Pass, Low Pass, No Pass
- **Course components:**
 - **Flipped classroom:** videos, readings, and daily pre-lecture problems
 - **Assignments:** Weekly assignments & Learning Process & Resource Reflection
 - **Exams:** Oral assessments, Computer-based exams
 - **Project:** Communication option, Kaggle option
- **Roughbooks:** Practice, practice, practice!

What is this course about?

- What is “learning” and why is our course “deep learning”?
 - Often about recognizing patterns
 - Deep learning: in a neural network architecture, many layers
- What’s in this course that you haven’t learned about in past ML/AI courses?
 - Overlap is good!
- What’s in this or any other syllabus that you would/wouldn’t expect?
- Some points of interest (instructor positionality):
 - An emphasis on data sourcing
 - Cognition

How to create something that “learns”

- Representations
 - An image could be a set of edges
 - A sound could be a composition of waves with different frequencies
- Architectures
 - Connections between cells in the brain
 - Connections between neural units

What skills should you have to create something that learns?

What do you think?

What skills **should** you have to create something that learns?

(As opposed to: what skills do you *need*)

- Math to envision it
- Programming to build it
- Communication to collaborate
 - You've chosen a field (and we exist in an era) in which it's very unlikely that you'll be sitting in an office building things alone!
- Ethics
 - Data: sourcing, labor
 - Architecture: handling biases
 - Deployment: access, environmental concerns

“That’s not my department!” says Wernher von Braun

Don't say that he's hypocritical
Say rather that he's apolitical
"Once the rockets are up, who cares where they come down?
That's not my department!" says Wernher von Braun.

- Tom Lehrer (“The Elements” song writer)

Goals: Learning & Credentialing

Learning: Understanding and appreciating the content

- Discussions in class
- Practice problems together and on your own
- Office hours

Credentialing/Certifying:

- Exams & proctored assessment

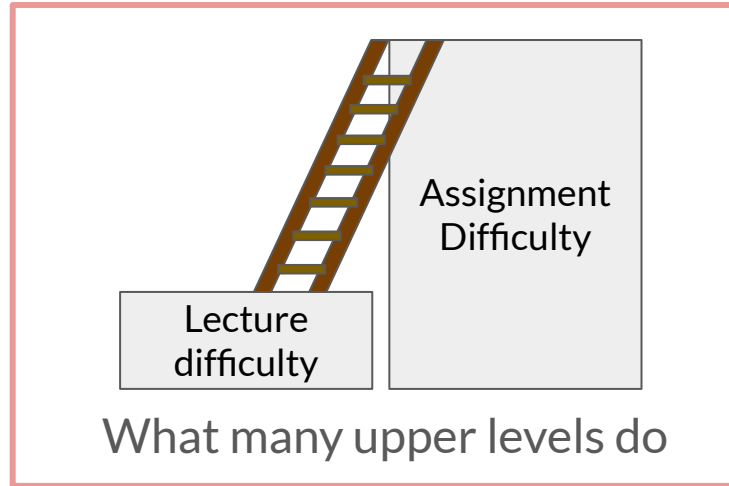
Sometimes these goals are hard to align.

When you have an “Aha!” moment, is there really an exam question that assesses that? Can you get an A in something you don’t understand?

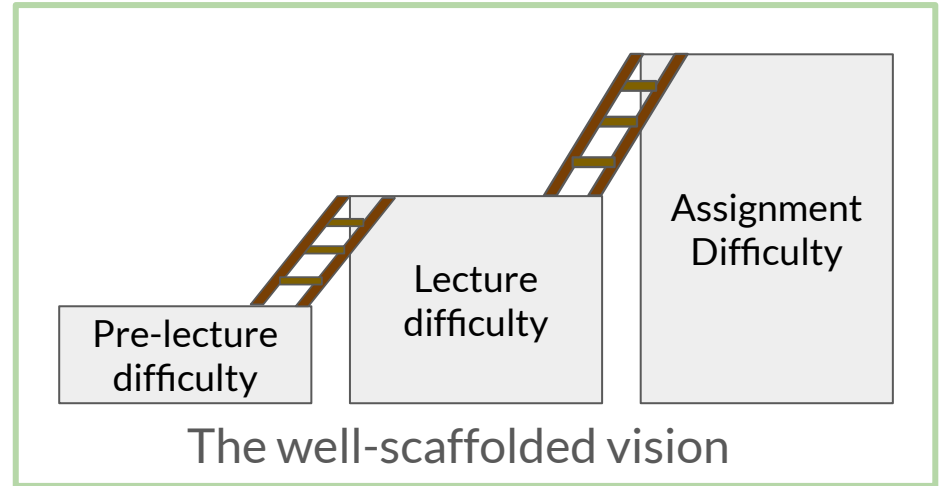
How to handle both learning and credentialing

- Learning: Flipped classroom
 - Understand/absorb content at your own pace
 - Lots of opportunities for discussion and problem solving
- Credentialing: In-person assessments with oral assessments and CBTF

Why a Flipped Classroom?



- Not very fun
- Equity: who has this big a ladder?



- More fun
- More equitable

(Theobald et al. 2020)

Standards-Based Grading

- For most components (other than CBTF exams), everything will be scored as:
 - High Pass (HP)
 - Pass (P)
 - Low Pass (LP)
 - No Pass (NP)
- To receive a particular letter grade, you must receive a certain threshold of Pass/High Pass for each course component (see syllabus)
- Questions?

A grade:

- > 80% on both exams
- High pass on at least 3 of 4 assignments, pass on remaining assignment
- High pass on all 4 of the Learning Process and Resource Reflections
- Completed 80% of pre-lecture homeworks for completeness before class
- High pass on 80% of pre-lecture homeworks for correctness, pass on remaining pre-lecture homeworks for correctness
- High pass on the project

A+: High pass on all components

A-: Meeting the B grade standards for exactly one of any of the above categories other than the exam score—a lower exam score moves to the B range

Standards-Based Grading

Helps with both **learning** and **credentialing**:

- Learning—care about the content, not every single “point”
- Credentialing—meet a certain threshold for each component; can’t “make up” for one competency with another

Weekly Tasks for the Course

- Pre-lecture homeworks: Monday, Tuesday, Wednesday
 - Watch videos and/or complete readings
 - Answer questions (for completeness) **by the night before**
 - Answer questions (for correctness) **by Friday night**
- Assignments: Due **Wednesday** night
 - Math-heavy portion
 - Programming-heavy portion
- Learning Process & Resource Reflection (LPRR): Due **Friday**—post on Piazza
- Oral exams on **Thursday** about the programming portion of your assignments.
Sign up by **Tuesday night** (Google Calendar sign up for this week will be available soon).
 - One “understand” question
 - One “experiment” question

Attendance

- Class attendance: mandatory, can miss up to 3 classes (out of 15 days of lecture)
 - Make a private Piazza post with your plan to catch up on the material missed during lecture
 - Submit on PrairieLearn a current version of the upcoming assignment (e.g. if you miss a day in week 2, submit your current A1) so that we can get a sense of how far along you are and if you're likely to be falling behind
- Discussion section: highly encouraged, and we expect you to **make a private post on Piazza to make a plan with us if you are planning to miss discussion regularly**. We'll be taking attendance just to get a sense of anyone who seems to be frequently missing

Learning Process & Resource Reflection

Think about all of the **resources that you really couldn't have done the assignment without** (GenAI, textbook, a Stackoverflow/Medium post or Github repo that you adapted your solution from). There may be multiple resources you used (e.g. maybe you used a particular Geeks4Geeks post a lot and also used GenAI a lot). Then, **pick one to focus on as you answer the following questions**, but feel free to mention if you used others along the way. The goal here is for you to communicate with your classmates and teaching staff how you're learning the course content—**what's working, and what isn't?**

We do ask that you not use GenAI for these—we really do want to know what you did, and there are no right or wrong answers, so there shouldn't be any motivation for you to use it if we all agree on the goal of knowledge sharing to help each other!

Learning Process & Resource Reflection

Please answer the following questions, and post your response on Piazza by Friday each week (300 words minimum):

1. Provide a link/brief description of at least **one** resource you felt was critical to your completion of this assignment. Feel free to list as many as you'd like (this question does not contribute to the word count).

(Then pick one to answer the remaining questions about).

2. How did you use the tool/resource at each step? For GenAI, were there particular prompts you used? For online resources, did you look at the solution and adapt it? Did you try other problems for practice after using the solution you found? No right or wrong answers—just be sure to explain your process! Whether you're satisfied or unsatisfied with the way you used available resources, explain your method (and you can reflect on whether it was helpful in the next question).
3. In what ways did this help and/or hinder your learning process? This is an important metacognitive activity regardless of GenAI—reflecting on your learning process can help both us and you reflect on what you're doing and whether or not it's helpful for meeting the intended learning objectives.
4. How did you decide whether to use the material or output from the tool/resource? Did you have particular test cases? Were there components in the solution that you were looking to make sure it included?
5. Provide clear recommendations for classmates on what parts of your process you found helpful and unhelpful.
6. Vote on your classmates' posts!

Oral Assessments: Motivation

- Anywhere you go next (industry, academia, nonprofit, government) will require you to communicate about what you build.
- Even if you use GenAI to write your code, you're responsible for it!
- If you had a coworker who brought code to you and had no idea what it was doing or how it was useful to the project, is that helpful?

Oral Assessments

10 minute oral assessments at a time slot you sign up for on Thursdays:

- “Understand” question: Explain to me what’s happening in this part of your code.
- “Experiment” question: If we were to change this about your code (e.g. change the data, some parameter, part of the architecture, etc.), what would change in how your code works and the output?

Oral Assessment Grading

- High Pass: Answer correctly with minimal prompting from teaching staff.
- Pass: Answer correctly with some prompting or correction from teaching staff.
- Low pass: Answer correctly with extensive prompting or correction from teaching staff.
- No pass: No-show or only incorrect answers.

You earn your submitted assignment grade by earning a High Pass on your oral assessment.

- Your grade for the assignment is determined by both the oral assessment and your autograder score. If you get a High Pass on the assignment but Low Pass on the oral assessment, you earn a Low Pass.

Each week, you can re-take one question from the previous week for a full replacement.

Exams

- Exams: Sign up through the TTC. Two days available for each exam.
 - Math & conceptual – assignments and practice exams
- Exam 1: Jul 9/10
 - Retake: Jul 13/14
- Exam 2: Jul 27/28
 - Retake: Jul 29/30
- Exam Retakes: 80% of your grade from the second attempt, 20% from the first

Project

Communication option

- Create an artifact (e.g. video, blog post, article, game, anything!) to explain or teach something in DL in a novel context
- Deliverable: **artifact** and **paper (experience report)**
- 10 minute presentation

Kaggle option

- Create a solution using any DL algorithms and architectures to score well on a given dataset
- Deliverable: **code** and **paper (research paper)**
- 3 minute lightning talk

Full details to come later in the week. Proposals are due next **Monday (Jul 6)**.

Experience Report

- Novel tool/system/strategy you tried
- How it went (“experience”)
- What people who used it thought overall
- Can inform future research—what should we invest in rigorously exploring?
- (Yours will not need to say how it went for people, though you’re welcome to!)

Research Paper

- Has research questions
- Methods & results answer research questions
- (Your RQs are likely “does this method hit these benchmarks”, but you could consider others!)

Read more: <https://sigcse2026.sigcse.org/track/sigcse-ts-2026-Papers>

(Note—your context does not have to be about education!)

Project Proposal : A structured abstract (250-350 words)

- **Background & Context**

- Communication: Who is your audience, what are you creating, why is it important, and what already exists? Why are existing videos, articles, etc. not sufficient for your context/audience?
- Kaggle: What are you solving, why is it important, and what has been tried already to solve it?

- **Goals**

- Communication: What do you intend to create?
- Kaggle: What are you trying to achieve? Likely benchmarking.

- **Methods**

- Communication—how you're making your design choices and creating your artifact
- Kaggle—what algorithms and architectures you're planning to explore

- **Results**

- Communication—explain your artifact, highlighting its affordances and drawbacks
- Kaggle—write what you hypothesize will happen if you use the methods you explain. Can suggest several possible hypotheses and outcomes.

- **Implications**

- Both: What are the implications?

Example Structured Abstract

Planning on Paper: Problem Decomposition with Diagrams in Introductory Computing

ANNAPURNA VADAPARTY, University of California, San Diego, USA

DEVAMARDEEP HAYATPUR, University of California, San Diego, USA

ADALBERT GERALD SOOSAI RAJ, University of California, San Diego, USA

LEO PORTER, University of California, San Diego, USA

DANIEL ZINGARO, University of Toronto Mississauga, Canada

Background and Context. Problem decomposition is a core concern of computing education. It has also become increasingly relevant: in response to GenAI, many CS1 educators are advocating for shifting instructional emphasis away from code writing and towards decomposition and higher-level planning. Currently, there is a lack of knowledge in how novices do decomposition in large, multifunction tasks.

Objectives. In this study, we describe how students represent solutions to a decomposition task, and characterize common issues that arise in those representations.

Method. In a 50-minute lab, students were given a description of a word game and asked to draw (with pencil and paper) a decomposition diagram for a program that would implement this game. We performed an inductive thematic analysis with negotiated agreement on 55 of the diagrams, coding salient elements (e.g. functions and the relationships between them) and issues that arose.

Findings. Students used multiple representational strategies, including hierarchical function calls and sequencing (order of execution). We identified issues in notation (including use of differing, incompatible notations within the same diagram), order of execution, abstraction and reuse, encapsulation, clarity, and problem-specific misunderstandings.

Implications. These findings suggest that novice decomposition is shaped by multiple underlying models of program behavior, with tensions between structural and sequence-focused reasoning. We discuss implications for decomposition instruction and future work, including clarifying representational constraints and plan tracing as simulation.

How to Submit

- Pre-Lecture homeworks for completeness: PrairieLearn
- Pre-Lecture homeworks for correctness: PrairieLearn
- Assignments math/conceptual: PrairieLearn
- Assignments programming: TBD
- Learning Process & Resource Reflection: Piazza Post
- Project Proposal: PrairieLearn
- Project: TBD

Exams: CBTF/TTC

(I use the phrases “computer based testing facility” and “Triton Testing Center” interchangeably.)

- Math portions
- Conceptual portions
- Assignments & practice exams will be good preparation

Late Hours

Late days & exceptions

Late time is counted hourly. Below are details for each type of submission.

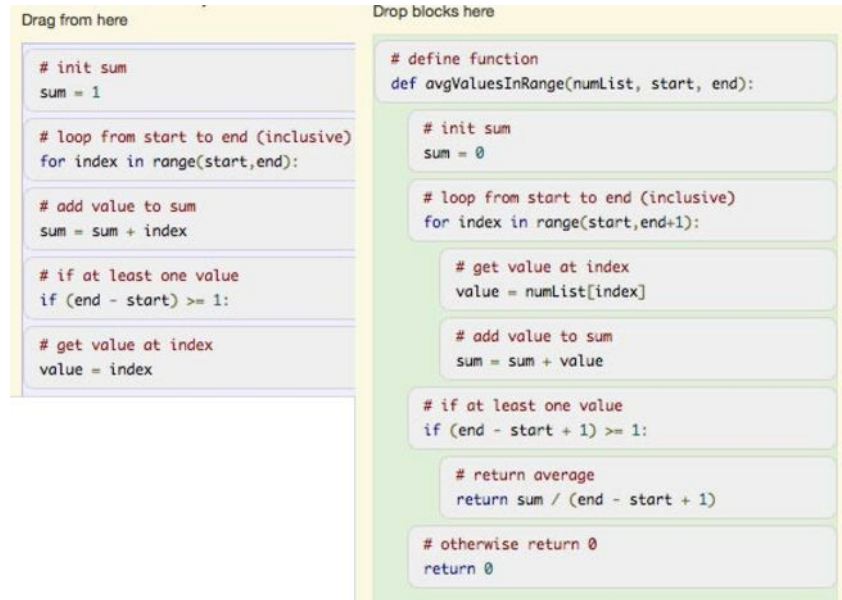
- **Assignments:** 48 late hours (2 late days) total for the course, max 12 late hours per assignment — so assignments are due Thursday at noon at the latest, to keep you on track for the Thursday oral assessment.
- **Pre-lecture homeworks (completeness):** due before class each day; 3 lowest dropped.
- **Pre-lecture homeworks (correctness):** 48 late hours total for the course, max 1 late day each; since these are all due Friday, the latest is Saturday night.
- **Project:** 24 late hours (1 late day) total across the proposal and the final submission.

Anything more than these times or extenuating circumstances: please contact Annapurna to discuss.

Assignments are Autograded

- Gives you immediate feedback to learn from
- Avoids grading time for teaching staff
- Free response converted to:
 - Parsons problems
 - Multiple choice
 - Fill in the blank

Parsons problems



The image shows a Parsons problem interface with two columns. The left column, labeled "Drag from here", contains five code blocks. The right column, labeled "Drop blocks here", contains the same five code blocks rearranged to form a complete Python function.

Drag from here

```
# init sum
sum = 1

# loop from start to end (inclusive)
for index in range(start,end):

# add value to sum
sum = sum + index

# if at least one value
if (end - start) >= 1:

# get value at index
value = index
```

Drop blocks here

```
# define function
def avgValuesInRange(numList, start, end):

# init sum
sum = 0

# loop from start to end (inclusive)
for index in range(start,end+1):

# get value at index
value = numList[index]

# add value to sum
sum = sum + value

# if at least one value
if (end - start + 1) >= 1:

# return average
return sum / (end - start + 1)

# otherwise return 0
return 0
```

But you should work on them yourself first!

- PDFs of open-ended versions of problems posted
- You can think of the PrairieLearn formulation (Parsons, MC) as scaffolding to help you out if you're not sure how to start
 - In PL make sure you scroll sideways for the Parsons problems – not everything shows in each block without scrolling
- Try them yourself first! (A0's on PL will open at 5pm)

Operationalizing “work on them yourself first”

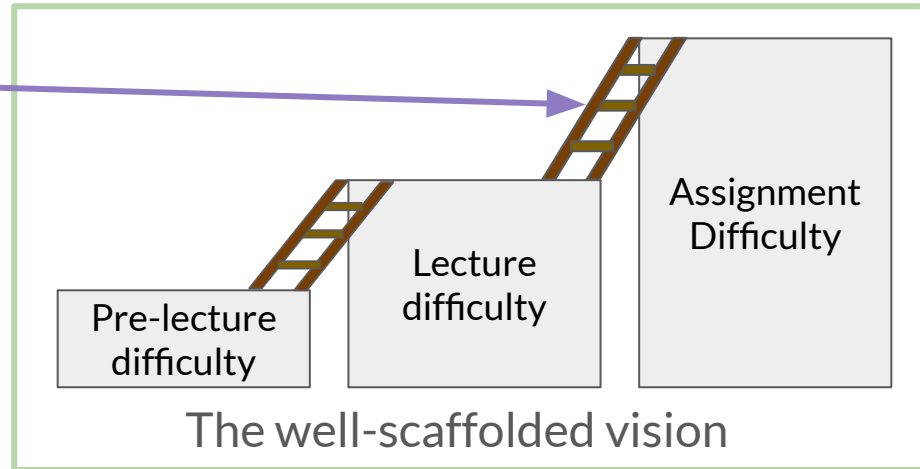
Doing problems yourself on paper is probably less frustrating anyway!

- You’ll feel like you know what’s going on rather than trying to figure out what each Parson’s block is trying to do (like trying to debug code you’re not familiar with).
- But don’t feel bad if you need to look! This probably just means that once you’re done, it’s a good idea to go find other practice problems like it.

Not Covered in Lecture/Pre-lecture? Find More Practice.

- There will be gaps between the pre-lecture homeworks and the assignments. I've done a lot to scaffold your learning, but it's an ongoing vision!
- We will post suggested resources and practice problems.
- This is a lot—your effort is appreciated! We aim for your total time to still be under the recommended credit hours.

This is still hard!

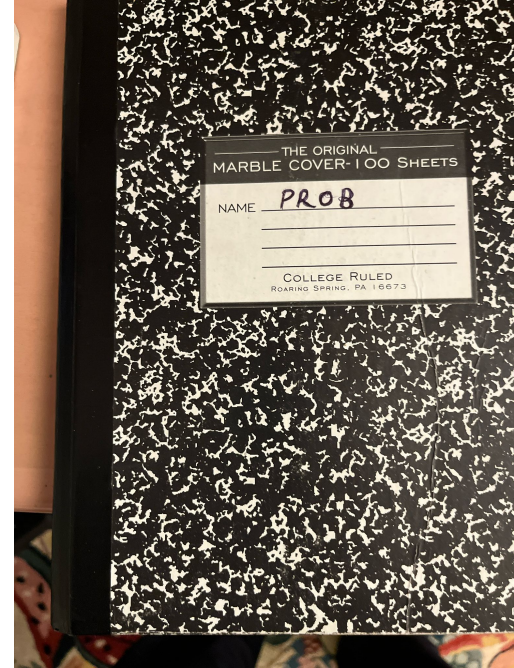
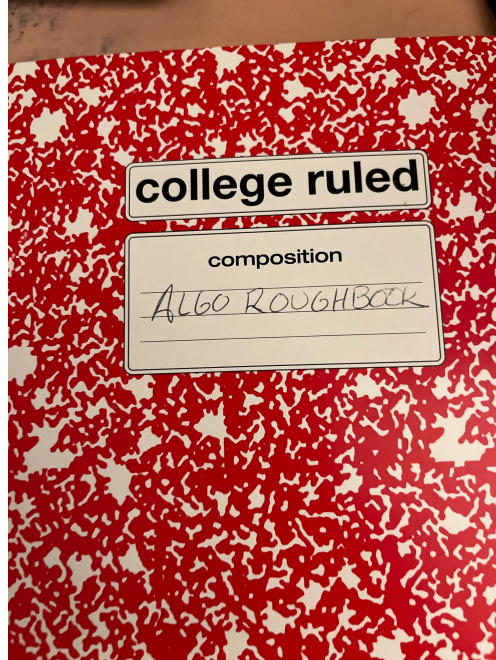


“Assignment Prep” Piazza Posts

- Assignment Prep Posts on Piazza: resources to help learn how to do questions like those on the assignments
- I’ll start off with some that I think are helpful, and I welcome you to comment with more you find! (Both math and programming resources welcome)
- This might also help you with your Learning Process & Resource Reflections

An Ode to Roughbooks

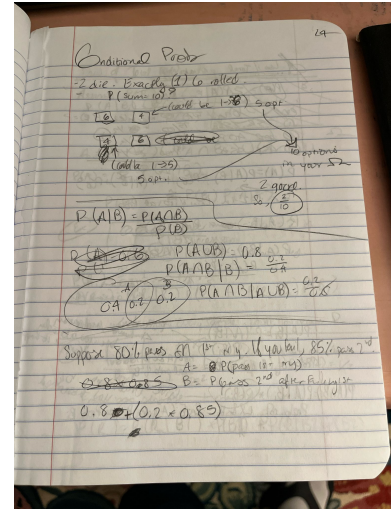
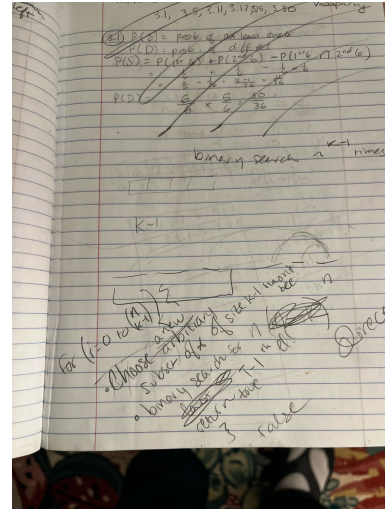
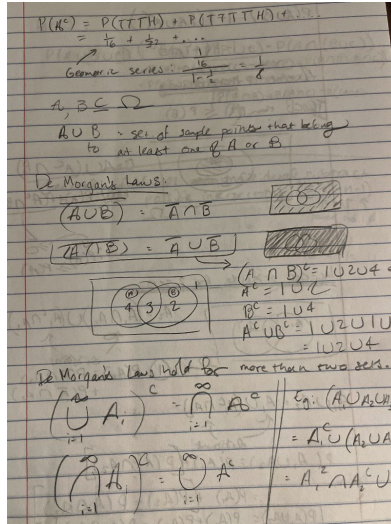
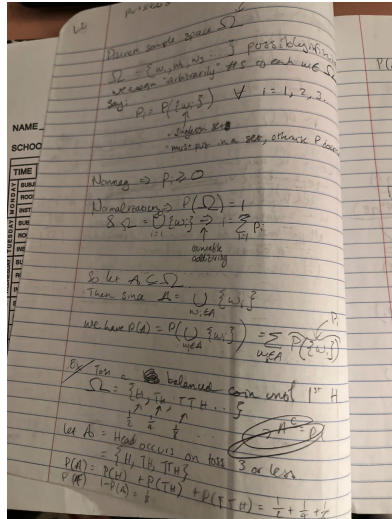
What is a roughbook,
and why did I get one for
all of you?



Roughbooks are...rough!

They make no promises about
handwriting...

Or correctness...



Their promise is to help you learn.

productive struggle



About 2,370,000 results (0.16 sec)

Productive struggle in middle school mathematics classrooms

HK Warshauer - Journal of Mathematics Teacher Education, 2015 - Springer

... This study suggests the **productive** role student **struggle** can play in supporting "... **Productive Struggle** Framework that I propose to capture the elements of a **productive** student **struggle** ...

☆ Save 📄 Cite Cited by 431 Related articles All 6 versions 🔗

Feeling Heard: Inclusive Education, Transformative Learning, and Productive Struggle

D Murdoch, AR English, A Hintz, K Tyson - Educational theory, 2020 - Wiley Online Library

... mean by affective dimensions of **productive struggle** and why these pose a risk for students. To do this, we delve into the philosophical roots of the concept of **productive struggle** and its ...

☆ Save 📄 Cite Cited by 130 Related articles All 6 versions 🔗

How Productive Is the Productive Struggle? Lessons Learned from a Scoping Review.

JR Young, D Bevan, M Sanders - ... of Education in Mathematics, Science and ..., 2024 - ERIC

... Together these definitions encompass many conceptualizations as well as actualizations of the **productive struggle**. In the present study we proffer that the **productive struggle** refers to ...

☆ Save 📄 Cite Cited by 44 Related articles All 3 versions 🔗

desirable difficulty



About 5,950,000 results (0.32 sec)

Desirable difficulties in theory and practice.

RA Bjork, EL Bjork - Journal of Applied research in Memory and ..., 2020 - psycnet.apa.org

... that our thinking about **desirable difficulties** needs to be ... less well when faced with **desirable difficulties**, how can that ... concepts or assumptions about **desirable difficulties**, the role played ...

☆ Save 📄 Cite Cited by 461 Related articles All 10 versions Web of Science: 176

[PDF] Making things hard on yourself, but in a good way: Creating desirable difficulties to enhance learning

EL Bjork, RA Bjork - Psychology and the real world: Essays ..., 2011 - responsetorisk.org

... My interests in the application of "**desirable difficulties**" were fanned by my experiences teaching and coaching and from what I learned as Chair of the National Research Council ...

☆ Save 📄 Cite Cited by 2237 Related articles All 14 versions 🔗

Worth the effort: The start and stick to desirable difficulties (S2D2) framework

ABH de Bruin, F Biwer, L Hui, E Onan, L David... - Educational Psychology ..., 2023 - Springer

... when engaging in **desirable difficulties**: the "Start and Stick to **Desirable Difficulties** (S2D2)" ... perceived learning in (dis)engagement in **desirable difficulties**, and (2) to review evidence ...

☆ Save 📄 Cite Cited by 105 Related articles All 8 versions Web of Science: 57

Desirable difficulty: theory and application of intentionally challenging learning

A Nelson, KL Elias - Medical Education, 2023 - Wiley Online Library

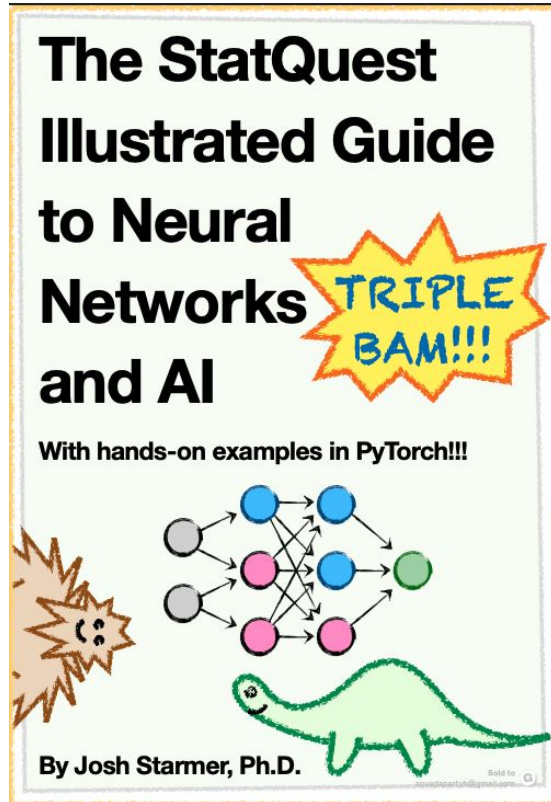
... **Desirable Difficulty** effect from three theoretical perspectives originating in different fields, discuss common evidence-based **Desirable Difficulty** ... could further optimise **Desirable Difficulty**-...

☆ Save 📄 Cite Cited by 55 Related articles All 5 versions Web of Science: 32 🔗

Flipped Classroom: 3Blue1Brown & StatQuest

- Every day there will be pre-lecture videos and/or readings
- Learn at your own pace and work through problems
- Submit on PL by the night before
- We'll take a look at your answers and see what people are most confused by to focus on in class the next day

StatQuest Textbook by Josh Starmer for everyone!



- The author was very kind and gave us a discount for his online book! I'm gifting you each a copy.
- Please be respectful of him and me—don't share/post. Credit and compensation for creators is important!
- UCSD emails—let me know if there is a different email you prefer.
- I will begin sending copies today afternoon once I get your pre-course survey response!

Questions for Teaching Staff?

Your TA and tutors are phenomenal!

Anything (their experience in this course, career thoughts, etc.) you'd want their perspective on?